Faculdade de Design,
Tecnologia e Comunicação
Universidade Europeia

02 – Data & Measurement Concepts

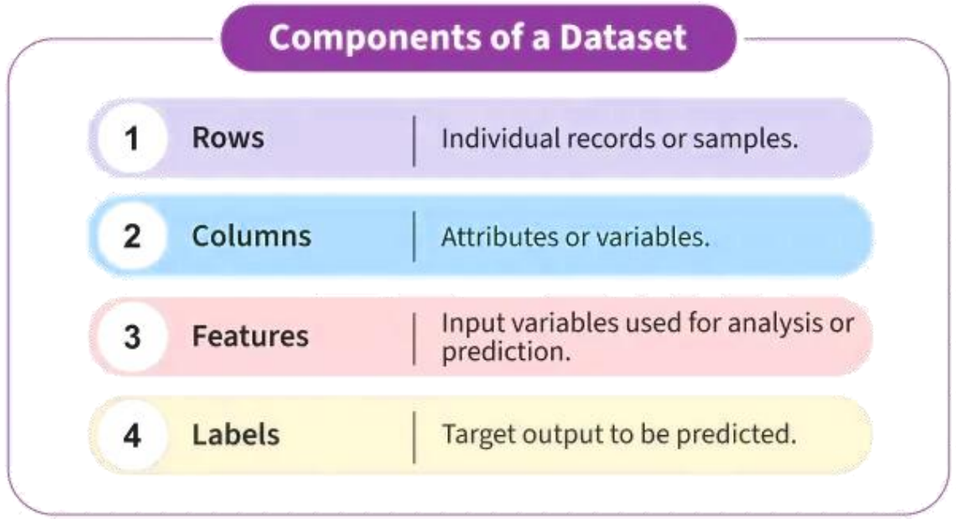
Automatic Applied Learning

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Dataset

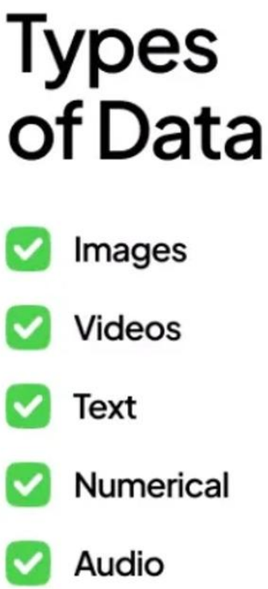
- A dataset is a **structured collection of data**, typically organized in a **tabular format** with **rows** and **columns**.



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Types of Dataset





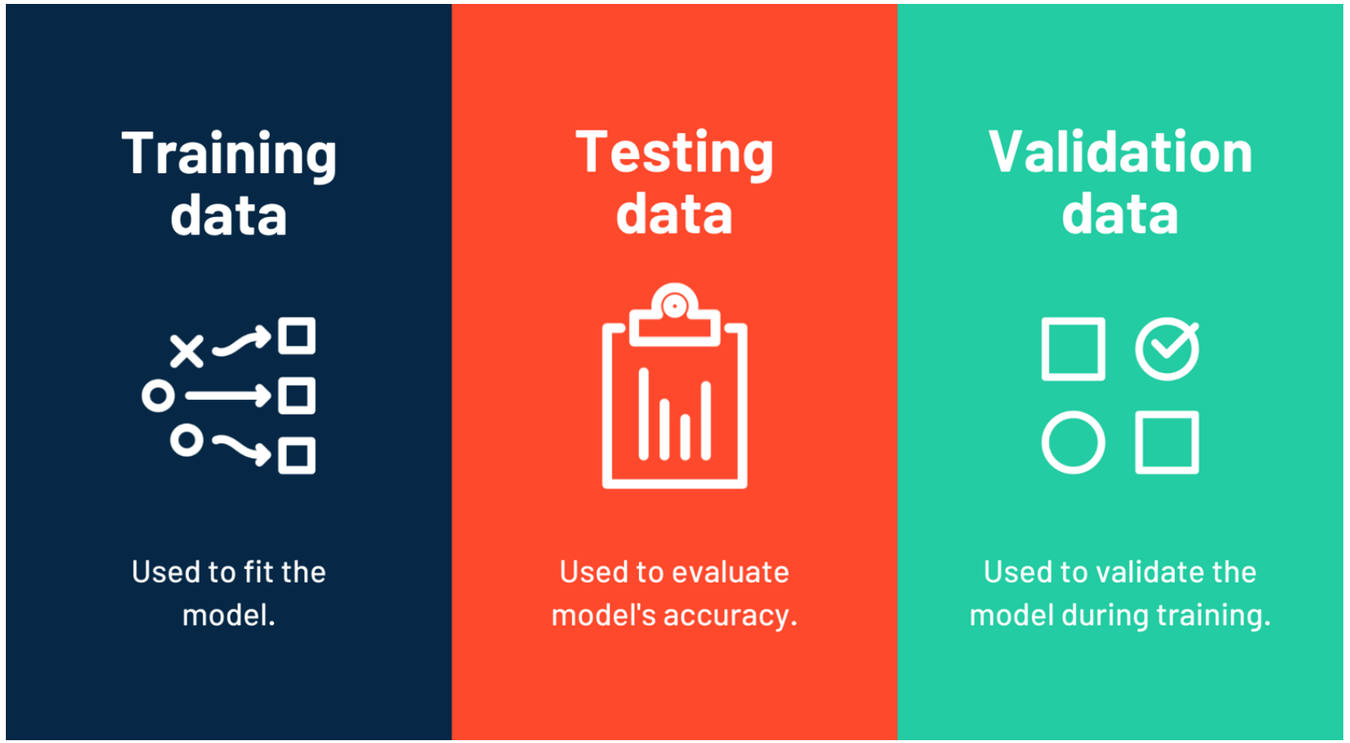
Sites

- https://www.kaggle.com/datasets
- **32 Best Free Datasets for Projects (2026)**

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Training, Validation, and Test Data

- A dataset is often divided into different parts.



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Training, Validation, and Test Data

A dataset is often divided into different parts.

Split	Purpose	Used for
Training	Fit model parameters	Learning weights, minimizing loss
Validation	Tune hyperparameters, compare models	Model selection, early stopping
Test	Final unbiased evaluation	Reporting generalization performance

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Generalization

A good model should work well with new examples, not only with the data used during training. This ability is called **generalization**.

Concepts

- **Training Error**: Error on training data (optimization objective)
- **Test Error**: Error on unseen data (true performance measure)
- **Generalization Gap**: Difference between test and training error

Example

- A model should recognize new photos of cats, not only remember the particular cat photos it saw during training.

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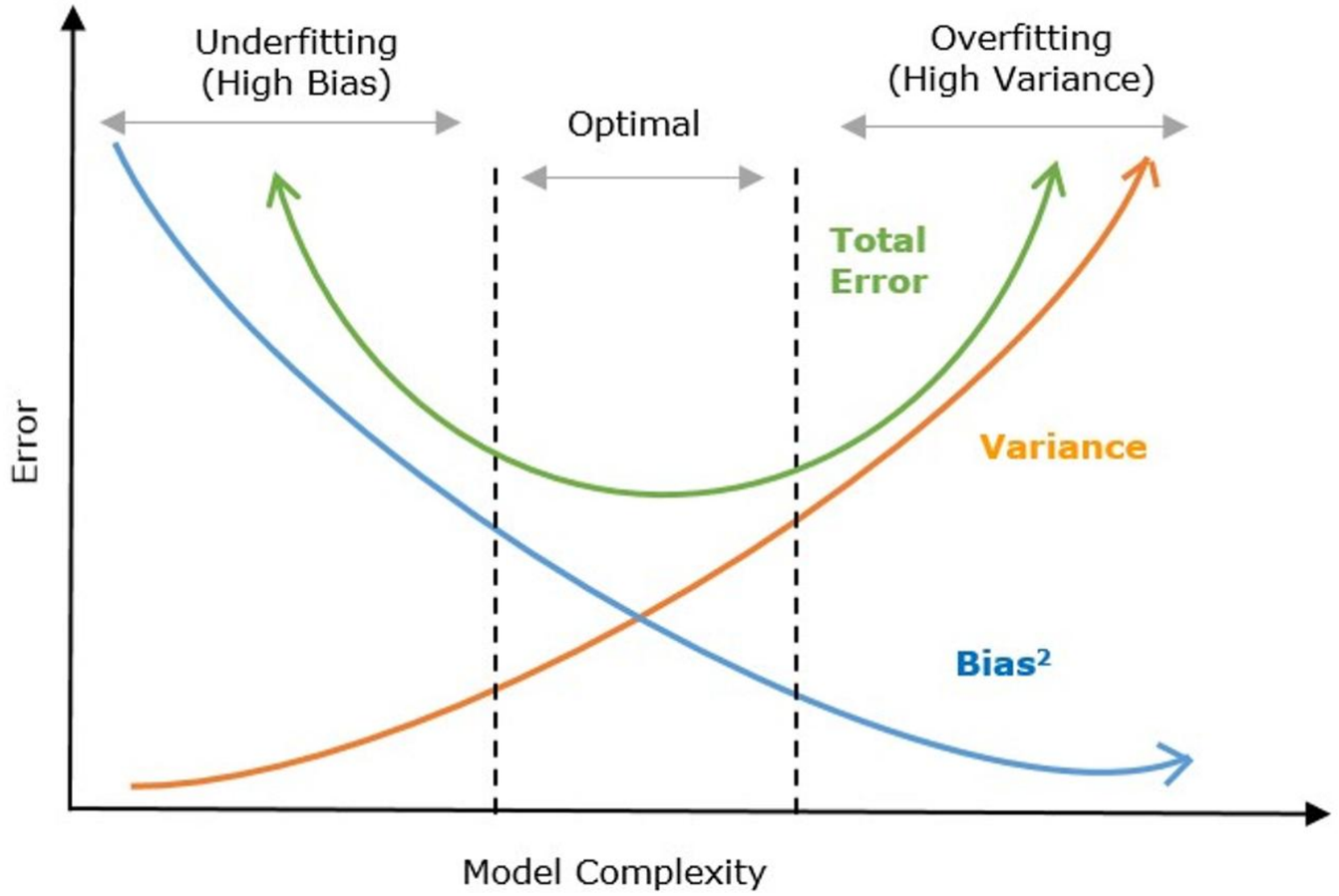
Generalization

- **Factors Affecting Generalization**

Factor	Impact on Generalization
Model Complexity	Too complex → overfitting; Too simple → underfitting
Training Data Size	More data → better generalization
Data Quality	Clean, representative data → better generalization
Regularization	L1/L2, dropout → reduces overfitting
Feature Engineering	Relevant features → better patterns

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Overfitting and Underfitting



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Overfitting and Underfitting

Overfitting

- Model learns training data too well, including noise.
- **Symptoms**: Low training error, high test error
- **Causes**: Too complex model, insufficient data, too many features
- **Solutions**: More data, regularization, simpler model, dropout, early stopping

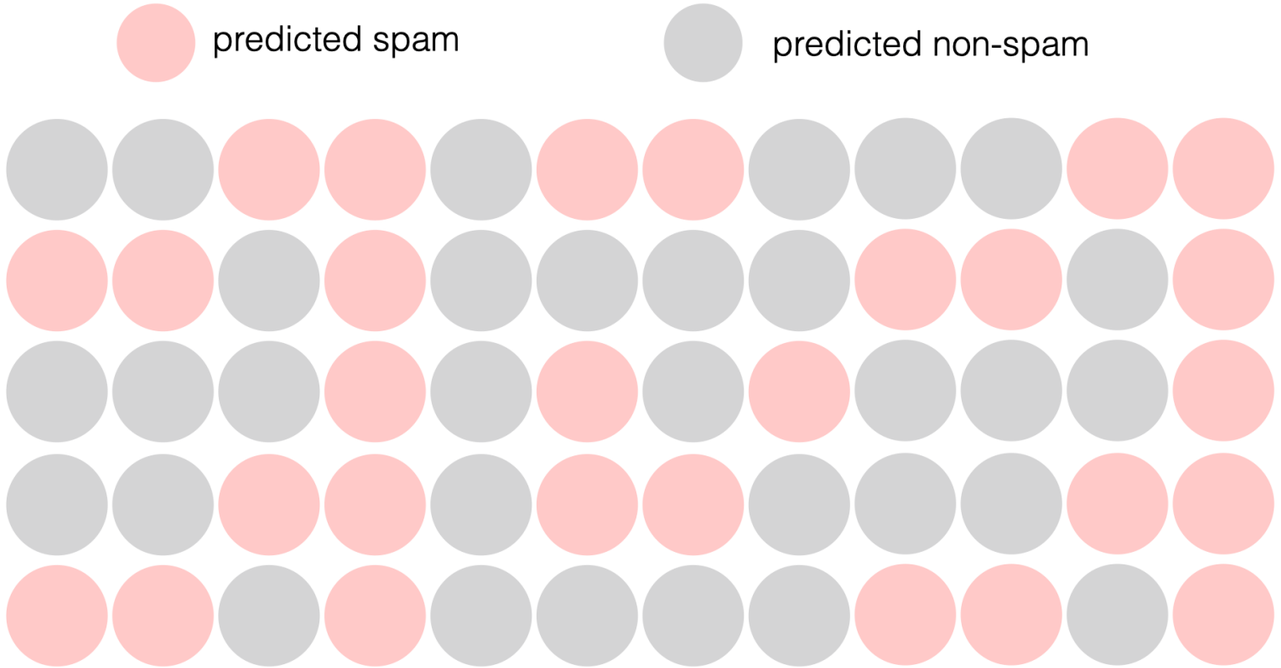
Underfitting

- Model fails to capture underlying patterns.
- **Symptoms**: High training error, high test error
- **Causes**: Too simple model, insufficient features, excessive regularization
- **Solutions**: More complex model, more features, reduce regularization, longer training

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Classification Metrics

- Accuracy
- Precision
- Recall
- F1 score



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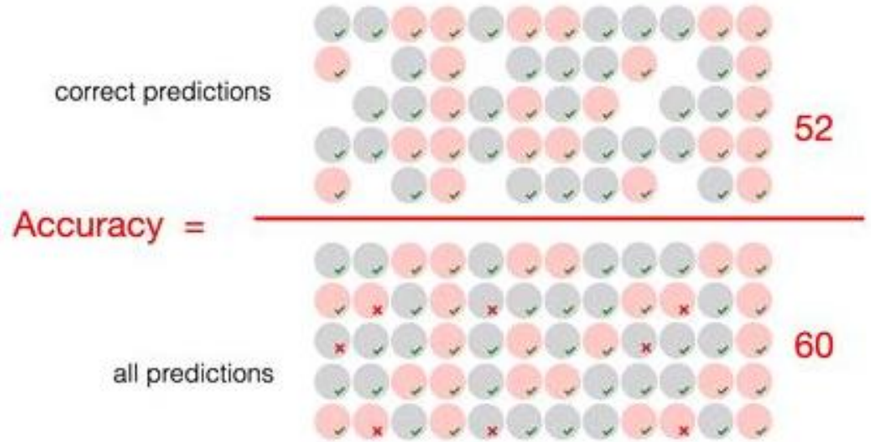
Accuracy

- **Concept**:

  The fraction of all predictions that were correct.

- **When to Use**: Balanced datasets, similar cost for **FP/FN**
- **Limitation**: Misleading for imbalanced data

  e.g., 99% negative → 99% accuracy by predicting all negative



$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

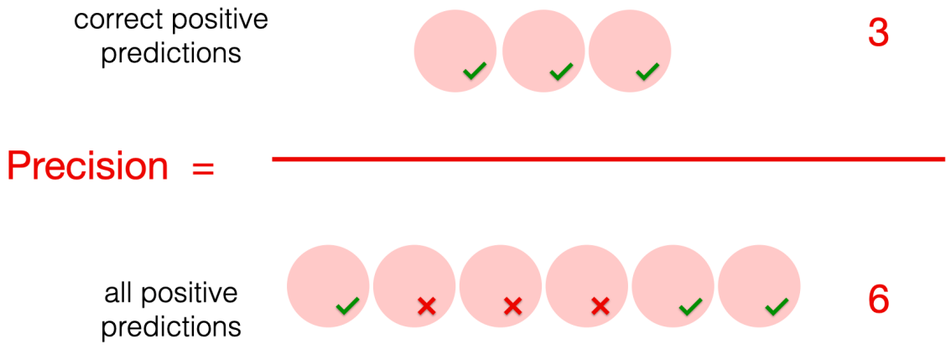
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Precision

- **Question**:

  Of all positive predictions, how many are correct?

- **When to Use**: Cost of FP is high
  - **Spam filtering** (don't mark legitimate as spam)
  - **Fraud alerts** (reduce unnecessary investigations)
  - **Medical diagnosis** (avoid invasive follow-ups for healthy patients)

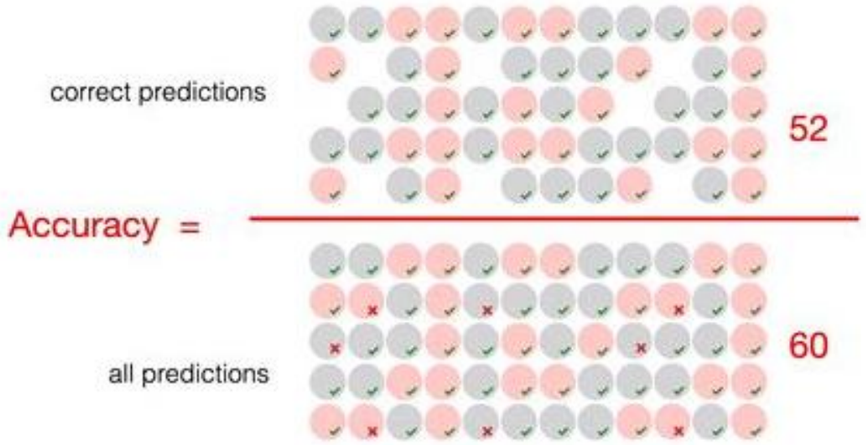


$$\text{Precision} = \frac{TP}{TP + FP}$$

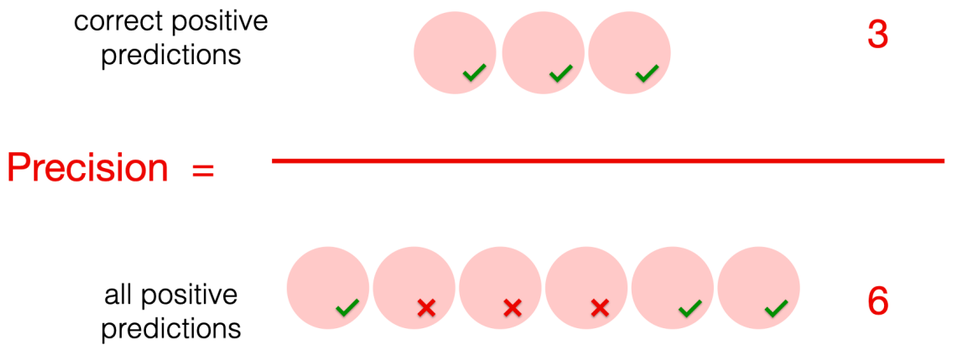
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Accuracy vs Precision

- **Accuracy** describes **how often a machine learning model makes correct predictions overall**, considering both **positive** and **negative cases**.



- **Precision** focuses **only on the model's positive predictions** and asks **how many of them were actually correct**.



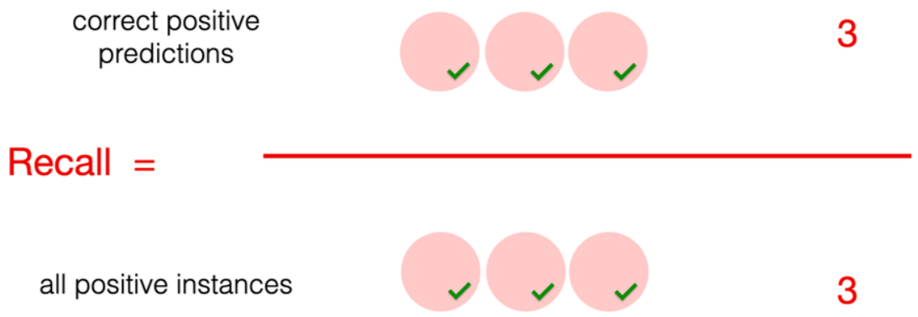
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Recall (Sensitivity / True Positive Rate)

- **Question**:

  Of **all actual positive cases**, how many did **we correctly identify as positive**?

- **When to Use**: Cost of **FN** is high
  - **Disease screening** (missing cases is dangerous)
  - **Fraud detection** (minimize undetected fraud)
  - **Safety systems** (don't miss critical events)



$$\text{Recall} = \frac{TP}{TP + FN}$$

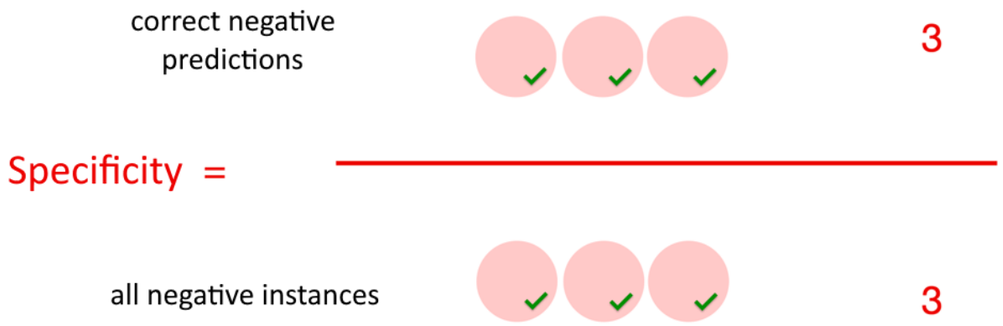
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Specificity (True Negative Rate)

- **Question**:

  Of **all actual negative cases**, how many did **we correctly identify as negative**?

- **When to Use**: Cost of **FP** is high
  - **Disease screening** (avoid unnecessary tests and anxiety)
  - **Fraud detection** (avoid blocking legitimate transactions)
  - **Spam filters** (avoid sending important messages to spam)



$$Specificity = \frac{TN}{TN + FP}$$

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F1 Score

- The F1 score combines **precision** and **recall to measure how well a model identifies positive cases** while **avoiding incorrect positive predictions.**

- **Properties**:
  - Harmonic mean of precision and recall
  - Range: [0, 1], higher is better
  - Balances precision and recall
  - Better than accuracy for imbalanced data

$$\mathrm{F1} = 2 \times \frac{\mathrm{Precision} \times \mathrm{Recall}}{\mathrm{Precision} + \mathrm{Recall}}$$

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Confusion Matrix

- **Confusion matrix** is a simple table used **to measure how well a classification model is performing**.
- It compares the **predictions made by the model** with the **actual results** and shows where the model was **right** or **wrong**.
- This helps you understand **where the model is making mistakes** so you can improve it.

	Actually Positive (1)	Actually Negative (0)
Predicted Positive (1)	True Positives (TPs)	False Positives (FPs)
Predicted Negative (0)	False Negatives (FNs)	True Negatives (TNs)

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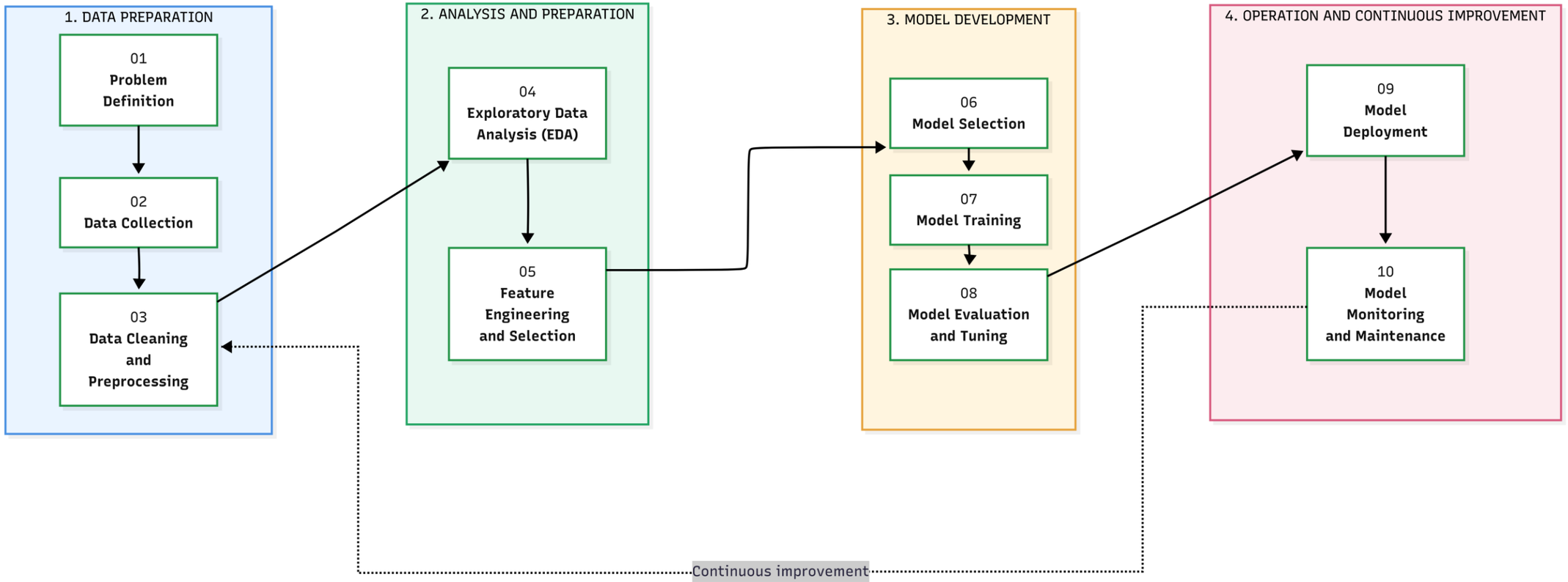
Confusion Matrix

- It breaks down the predictions into four categories:
  - **True Positive (TP):** The model correctly predicted a positive outcome i.e the actual outcome was positive.
  - **True Negative (TN):** The model correctly predicted a negative outcome i.e the actual outcome was negative.
  - **False Positive (FP):** The model incorrectly predicted a positive outcome i.e the actual outcome was negative. It is also known as a Type I error.
  - **False Negative (FN):** The model incorrectly predicted a negative outcome i.e the actual outcome was positive. It is also known as a Type II error.

	Actually Positive (1)	Actually Negative (0)
Predicted Positive (1)	True Positives (TPs)	False Positives (FPs)
Predicted Negative (0)	False Negatives (FNs)	True Negatives (TNs)

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Machine Learning Lifecycle



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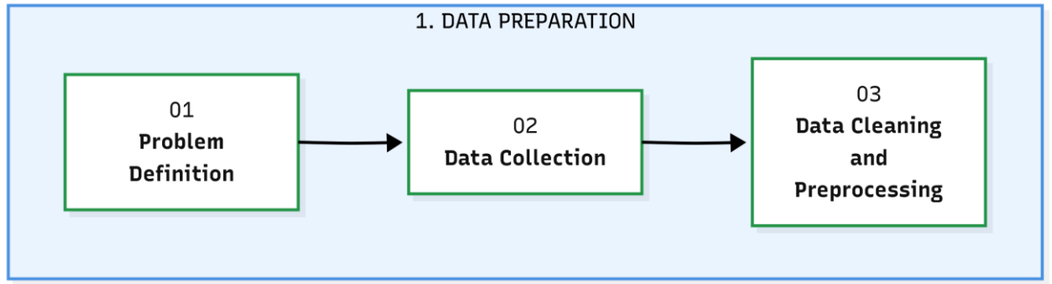
Machine Learning Lifecycle

Phase	Stage	Group	Description
01	Problem Definition	Data Preparation	Define the problem, objectives, constraints, and success criteria.
02	Data Collection	Data Preparation	Gather relevant data from reliable sources.
03	Data Cleaning and Preprocessing	Data Preparation	Correct errors, handle missing values, remove duplicates, and prepare the data for analysis.
04	Exploratory Data Analysis (EDA)	Analysis and Preparation	Analyze patterns, relationships, distributions, and possible anomalies in the data.
05	Feature Engineering and Selection	Analysis and Preparation	Create, transform, and select the most useful features for the model.
06	Model Selection	Model Development	Choose an appropriate algorithm based on the problem and data characteristics.
07	Model Training	Model Development	Train the selected model using the prepared data.
08	Model Evaluation and Tuning	Model Development	Measure performance and optimize the model's parameters.
09	Model Deployment	Operation and Continuous Improvement	Make the trained model available for use in a production environment.
10	Model Monitoring and Maintenance	Operation and Continuous Improvement	Track performance, detect issues, and update the model when necessary.

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Machine Learning Lifecycle

Data Preparation

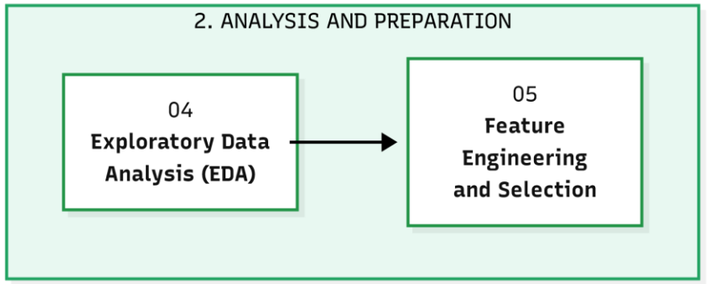


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Machine Learning Lifecycle

Analysis and Preparation

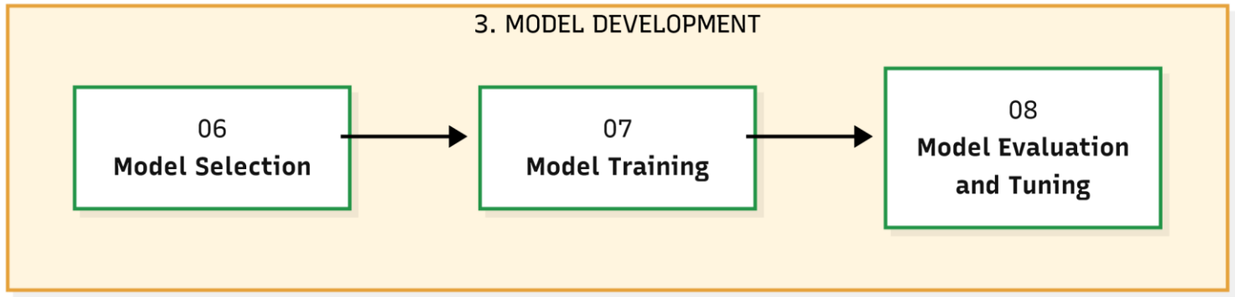


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Machine Learning Lifecycle

Model Development

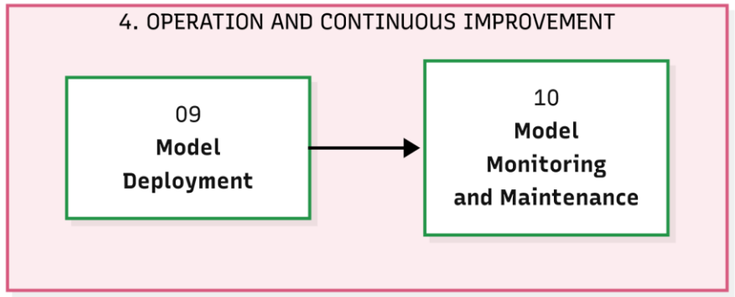


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Machine Learning Lifecycle

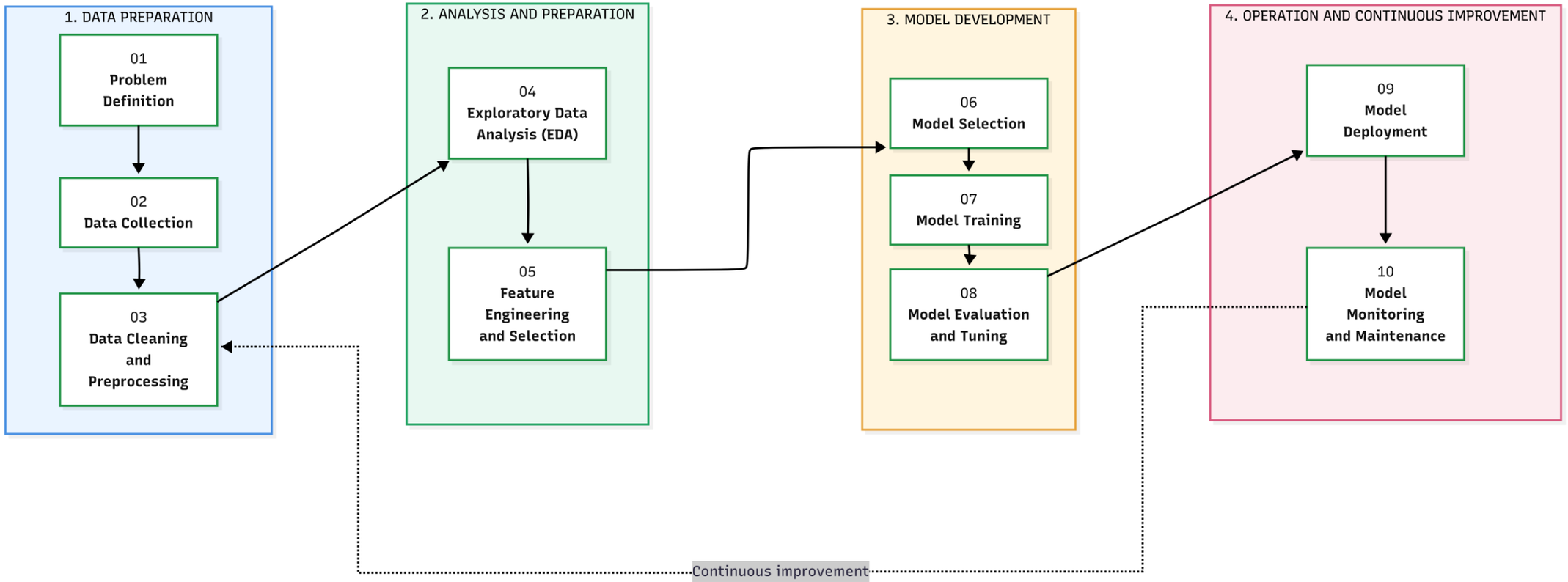
Operation and Continuous Improvement



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Machine Learning Lifecycle



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